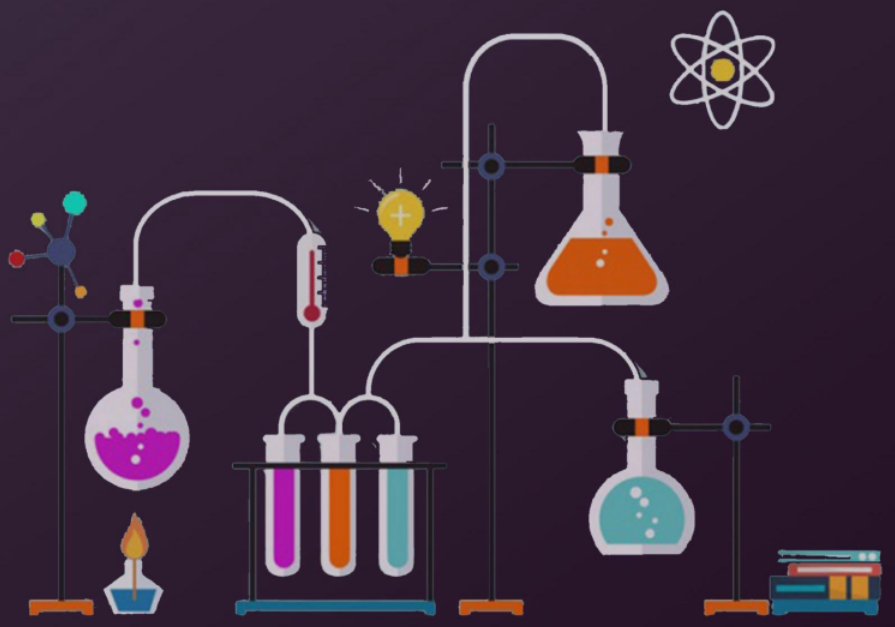
CHEMISTRY

Electrochemistry



SHEET

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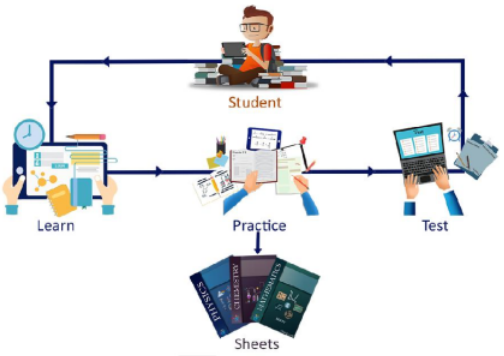
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction and Galvanic Cell	1			
2	Representation of Galvanic Cell and Eletcrode Potential Measurement	2-6			
3	Relationship between Chemical and Electrical Energy in a Cell	7-12		1-11	8,9,21
4	Nerst Equation and Concentration cells	13-16	1,4-8,19, 24-26	12-32	1,10-17, 23-25
5	Electrolysis and its products	17-19	9-11	33-42	2,19,26
6	Faraday's law of Electrolysis	20-27	2,17-18,20, 21,27-30	43-44	
7	Conductance of an Electrolytic Solution	28-34	3, 12-15	45-48	3
8	Effect of Temperature and Dilution on Conductance	35-40	16,22,23, 31-34	49-52	4,5,20, 22,27,28
9	Ionic Mobility and Conductometric Titration	41-42			
10	Commercial Batteries				6,7
11	Corrosion	43-44			
12	ACC	45-46			18

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction and Galvanic Cell

Q1. If the emf of a galvanic cell is negative, it implies that :-

(a) The cell reaction is spontaneous

(b) The cell reaction is non-spontaneous

(c) The cell reaction is exothermic

(d) The cell is working in reverse direction

Correct code is?

(A) a, b, d (B) b, c, d

(C) b, d (D) a, c, d

Representation of Galvanic cell And Electrode Potential Measurment

Q2. The chlorate ion can disproportinate in basic according to reaction $2ClO^-_3 \rightarrow ClO^-_2 + ClO^-_4$ calculate the value of equilibrium constant given Given $E^\circ_{ClO^-_4/ClO^-_3} = 0.36V$ and $E^\circ_{ClO^-_3/ClO^-_2} = 0.33V$ at 298 K :-

(A) 0.1 (B) 0.2

(C) 0.3 (D) 0.4

Q3. A solution is one molar in each of NaCl, $CdCl_2$, $ZnCl_2$ and $PbCl_2$. To this, tin metal is added.

Which of the following is true ? Given :-

$E^\circ_{Pb^{2+}/Pb} = -0.126$ V, $E^\circ_{Sn^{2+}/Sn} = -0.136$ V, $E^\circ_{Cd^{2+}/Cd} = -0.40$ V, $E^\circ_{Zn^{2+}/zn} = -0.763$ V, $E^\circ_{Na^+/Na} = -2.71$ V,

(A) Sn can reduce Na^+ to Na

(B) Sn can reduce Zn^{2+} to Zn

(C) Sn can reduce Cd^{2+} to

(D) Sn can reduce Pb^{2+} to Pb

Q4. Standard oxidation electrode potentials of four metals P,Q,R and S are +2.87, +3.05 – 0.80 and +0.25 volts respectively. The reducing power of these metals are ?

(A) P > Q > R > S

(B) S > R > Q > P

(C) Q > P > S > R

(D) None of these

Q5. In the cell $Zn|Zn^{2+}||Cu^{2+}|Cu$, the negative electrode is?

(A) Cu (B) Cu^{2+}

(C) Zn (D) Zn^{2+}

Q6. Calculate the oxidation potential of hydrogen electrode at pH = 10 & $P_{H_2} = 1$ atm :-

(A) –0.59 V (B) 0.00 V

(C) +0.59 V (D) 0.059 V

Relationship Between Chemical and Electrical Energy In a Cell

Q7. E° for the half cell $Zn^{2+}|Zn$ is – 0.76 volt. E.M.F. of the cell

$Zn_{(s)}|Zn^{2+}_{(aq)}(1M)||2H^+_{(aq)}(1M)|H_{2(g)}(1\text{ atm})$ is?

(A) – 0.76 V (B) + 0.76 V

(C) – 0.38 V (D) + 0.38 V

Q8. The standard cell potential of the galvanic cell $Zn_{(s)}|Zn^{2+}_{(aq)}||Ag^+_{(aq)}|Ag_{(s)}$, where $E^\circ_{Zn^{+2}/Zn} = -0.76$ V; $E^\circ_{Ag^+/Ag} = +0.80$ V:

(A) 0.04 V (B) –0.04 V

(C) 1.56 V (D) –1.56 V

Q9. If $E^\circ_{Au+/Au}$ is 1.69 V and $E^\circ_{Au+3/Au}$ is 1.40 V, then $E^\circ_{Au+/Au+3}$ will be ?

(A) 0.19 V (B) 2.945 V

(C) 1.255 V (D) -1.255 V

Q10. The thermodynamic efficiency of cell is given by-

(A) $\frac{\Delta H}{\Delta G}$ (B) $\frac{nFE_{cell}}{\Delta G}$

(C) $-\frac{nFE_{cell}}{\Delta H}$ (D) Zero

Q11. E° for the reaction, $Fe_{(s)} + Zn^{2+}_{(aq)} \rightarrow Zn_{(s)} + Fe^{2+}_{(aq)}$ is – 0.35V. The given cell reaction is ?

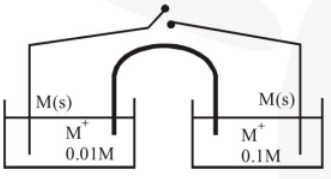
(A) spontaneous
(B) non spontaneous
(C) in equilibrium
(D) can't say anything

Q12. In a galvanic cell, the electrical work done is equal to -

(A) Free energy change
(B) Mechanical work done
(C) Thermodynamic work done
(D) All of these

Nernst Equation And Concentration Cells

Q13.



Initial EMF of above cell at 298K is?

(A) –0.059 V (B) 0.059 V
(C) 0.59 V (D) 0.0059 V

Q14. For a redox reaction, Oxi + $ne^- \longrightarrow$ red, the Nernst equation has a form of :

(A) $E_{cell} = E^o_{cell} + \frac{RT}{nF}\ell n\frac{[red]}{[Oxi]}$

(B) $E_{cell} = E^o_{cell} - \frac{RT}{nF}\ell n\frac{[red]}{[Oxi]}$

(C) $E_{cell} = E^o_{cell} - \frac{RT}{nF}\log\frac{[red]}{[Oxi]}$

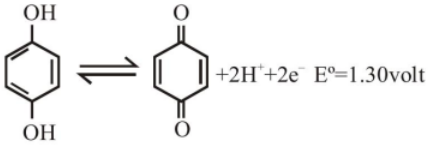
(D) $E_{cell} = E^o_{cell} + \frac{RT}{nF}\log\frac{[red]}{[Oxi]}$

Q15. What is the SRP for $S^{-2}|Cus|Cu$ half cell :-

(Given $K_{sp}(CuS) = 10^{-35}$ and $E^\circ_{Cu^{+2}/Cu} = 0.34$ V

(A) 1.034 V (B) 1.0 V
(C) – 0.694 V (D) 0.694 V

Q16. The cell reaction involving quinhydrone electrode is :-



What will be the electrode potential at pH = 3 ?

(A) 1.20 V (B) 1.10 V
(C) 1.48 V (D) 1.30 V

Electrolysis and It's Products

Q17. During electrolysis of fused calcium hydride (CaH_2), the hydrogen is produced at ?

(A) Cathode
(B) Anode
(C) Hydrogen is not liberated at all
(D) H_2 produced reacts with oxygen to form water

Q18. Which of the following arrangement will produced oxygen at anode during electrolysis:-

(a) Dilute H_2SO_4 with Pt electrode
(b) molten CaH_2 with inert electrode
(c) Fused NaOH with inert electrode
(d) Concentrate aq NaCl with Pt electrode

Correct code is ?

(A) a, c, d (B) a, c
(C) b, c, d (D) b, d

Q19. The passage of current liberates H_2 at cathode and Cl_2 at anode.The solution is?

(A) $CuSO_4$ (aq) (B) $CuCl_2$ (aq.)
(C) NaCl(aq.) (D) Water

Faraday's Law of Electrolysis

Q20. The amount of an ion discharged during electrolysis is not directly proportional to :

(A) resistance

(B) time

(C) current strength

(D) electrochemical equivalent of the element

Q21. Which one is the correct equation that represents the first law of electrolysis ?

(m → mass, c → current, t → time)

(A) mz = ct (B) m = czt

(C) mc = zt (D) c = mzt

Q22. Number of faraday required to liberate 8 gm. of H_2 is ?

(A) 8 (B) 16

(C) 4 (D) 2

Q23. How many minutes will it take to plate out 5.0g of Cr from a $Cr_2(SO_4)_3$ solution using a current of 1.50 A ?

(Atomic weight : Cr = 52.0)?

(A) 254 (B) 309

(C) 152 (D) 103

Q24. If current (I) of an electrolytic cell is doubled and time (t) for electrolysis is halved then $\frac{\text{mass}}{\text{eq.wt.}}$ ratio becomes :-

(A) doubled (B) halved

(C) 1.5 times (D) remain same

Q25. In passing 3 Faraday of electricity through the three electrolytic cells connected in series containing molten Ag^+, Ca^{+2} and Al^{+3} ions respectively the molar ratio in which the three metal ions are liberated at the electrodes is ?

(A) 1 : 2 : 3 (B) 3 : 2 : 1

(C) 6 : 3 : 2 (D) 3 : 4 : 2

Q26. Faraday's laws of electro,elated to the:

(A) atomic number of the cation

(B) atomic number of the anion

(C) equivalent weight of the electrolyte

(D) speed of the cation

Q27. Two electrolytic cells one containing acidified ferrous chloride and another acidified ferric chloride are connected in series. The ratio of mass of iron deposited at cathodes in the two cells when electricity is passed through the cells will be ?

(A) 3 : 1 (B) 2 : 1

(C) 1 : 1 (D) 3 : 2

Conductance of an Electrolytic Solution

Q28. Strong electrolyte are those which :

(A) dissolve readily in water

(B) conduct electricity

(C) dissociate into ions even at high concentration

(D) dissociate into ions at high dilution.

Q29. Electrolytic conduction is due to the movement of ?

(A) molecules (B) atoms

(C) ions (D) electrons

Q30. The conductivity of a saturated solution of Ag_3PO_4 is 9×10^{-6} S m^{-1} and its equivalent conductivity is 1.50×10^{-4} S m^2 equivalent^{-1}. The K_{sp} of Ag_3PO_4 is?

(A) 4.32×10^{-18} (B) 1.8×10^{-9}

(C) 8.64×10^{-13} (D) None of these

Q31. The resistance of 0.05 M solution of oxalic acid is 200 ohm and cell constant is 2.0 cm^{-1}, the equivalent conductance (in S cm^2 eq^{-1}) of 0.05 M oxalic acid is ?

(A) 100 (B) 10

(C) 1 (D) 0.1

Q32. Which of the following solution of KCl has the lowest value of specific conductance :

(A) 1 M (B) 0.1 M

(C) 0.01 M (D) 0.001 M

Q33. The specific conductance of a salt solution of 0.01M concentration is 1.061×10^{-4}. Molar conductance of the solution will be ?

(A) 1.061×10^{-4} (B) 1.061

(C) 10.61 (D) 106.1

Q34. The dissociation constant of n-butyric acid is 1.6×10^{-5} and the molar conductivity at infinite dilution is 380×10^{-4} Sm^2mol^{-1}. The specific conductance of the 0.01 M acid solution is ?

(A) 1.52×10^{-5} Sm^{-1} (B) 1.52×10^{-2} Sm^{-1}

(C) 1.52×10^{-3} Sm^{-1} (D) 1.52×10^{-4} Sm^{-1}

Effect of Temperature and Dilution on Conductance

Q35. The limiting molar conductivities $\wedge^\circ$ for NaCl, KBr and KCl are 126, 152 and 150 Scm^2mol^{-1} respectively. The $\wedge^\circ$ for NaBr is?

(A) 278 S cm^2mol^{-1}

(B) 176 S cm^2mol^{-1}

(C) 128 S cm^2mol^{-1}

(D) 302 S cm^2mol^{-1}

Q36. According to Kohlrasuch law, the limiting value of molar conductivity of an electrolyte A_2B is?

(A) $\lambda^\infty_{A^+} + \lambda^\infty_{B^{-2}}$ (B) $\lambda^\infty_{A^{+1}} - \lambda^\infty_{B^{-2}}$

(C) $\lambda^\infty_{A^+} + \frac{1}{2}\lambda^\infty_{B^{-2}}$ (D) $2\lambda^\infty_{A^+} + \lambda^\infty_{B^{-2}}$

Q37. 0.05 M weak acid HA was found to have molar conductivity of 4.0 S cm^2 mol^{-1} while that of HA at infinite dilution is 400 S cm^2 mol^{-1}. The degree of dissociation of 0.05 M acid is ?

(A) 2% (B) 2.3%

(C) 1% (D) 1.5%

Q38. The value of λ^∞_m for NH_4Cl, NaOH and NaCl are 129.8, 248.1 and 126.4 ohm^{-1} cm^2 mol^{-1} respectively. Calculate λ^∞_m for NH_4OH solution (in ohm^{-1} cm^2 mol^{-1})

(A) 215.5 (B) 251.5

(C) 244.7 (D) 351.5

Q39. The equivalent conductivity of H_2SO_4 at infinite dilution is 384 Ω^{-1} cm^2 eq^{-1}. If 49 gm H_2SO_4 per litre is present in solution and specific resistance is 18.4 Ω–cm then, calculate the degree of dissociation :-

(A) 11% (B) 12%

(C) 13%% (D) 14%

Q40. For saturated solution of CaF_2, choose the correct relationship :

(A) $\Lambda^\infty_{m_{CaF_2}} = \lambda^\infty_{eq_{Ca^{2+}}} + 2\lambda^\infty_{eq_{F^\ominus}}$

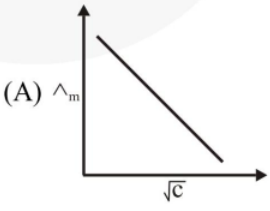
(B) $\Lambda^\infty_{m_{CaF_2}} = 2\left(\lambda^\infty_{eq_{Ca^{2+}}} + \lambda^\infty_{eq_{F^\ominus}}\right)$

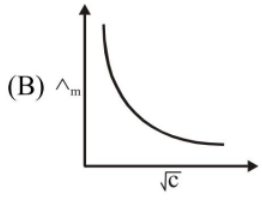
(C) $\Lambda^\infty_{m_{CaF_2}} = 2\left(\lambda^\infty_{eq_{Ca^{2+}}} + 2\lambda^\infty_{eq_{F^\ominus}}\right)$

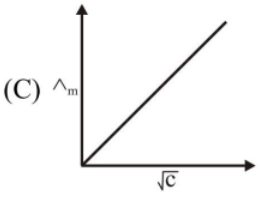
(D) $\Lambda^\infty_{m_{CaF_2}} = \frac{1}{2}\left(\lambda^\infty_{eq_{Ca^{2+}}} + \lambda^\infty_{eq_{F^\ominus}}\right)$

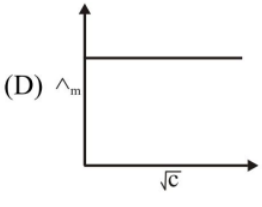
Ionic Mobility and Conductometric Titration

Q41. The variation of molar conductance of a weak electrolyte with concentration is shown in :-

(A) 

(B) 

(C) 

(D) 

Q42. For strong electrolyte $\Lambda_m = \Lambda_m^\circ - 10\sqrt{C}$. Then calculate Λ_m at 0.01M if its value at 0.16M is 200 $Scm^2\ mol^{-1}$.

(A) 203 $Scm^2\ mol^{-1}$

(B) 193 $Scm^2\ mol^{-1}$

(C) 207 $Scm^2\ mol^{-1}$

(D) 197 $Scm^2\ mol^{-1}$

Corrosion

Q43. Which of the following is an example of corrosion?

(A) Rusting of iron

(B) Tarnishing of silver

(C) Liquefaction of ammonia

(D) Rusting of iron and tarnishing of silver

Q44. Rusting is a special term, given to the corrosion of which metal?

(A) Zinc (B) Copper

(C) Nickel (D) Iron

ACC

Q45. Which aqueous solution can not be stirred with silver spoon-

(A) $CuSO_4$ (B) $Zn(NO_3)_2$

(C) $AuCl_3$ (D) $NiSO_4$

Q46. The oxide which can be reduced by hydrogen is ?

(A) Na_2O (B) CaO

(C) K_2O (D) CuO

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Any redox reaction would occur spontaneously, if ?

(A) the free energy change (ΔG) is negative

(B) the (ΔG) is positive

(C) the cell e.m.f. (E°) is negative

(D) the cell e.m.f. is positive

Q2. A current of 2.68 A is passed for one hour through an aqueous solution of $CuSO_4$ using copper electrodes. Select the correct statement (s) from the following :-

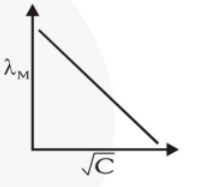
(A) increase in mass of cathode = 3.174 g

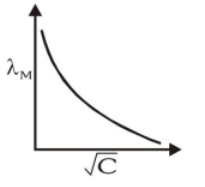
(B) decrease in mass of anode = 3.174 g

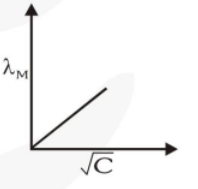
(C) no change in masses of electrodes

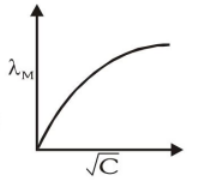
(D) the ratio between the change of masses of cathode and anode is 1 : 2

Q3. Which of the following curve represents the variation of λ_M with $\sqrt{C}$ for $AgNO_3$?

(A) 

(B) 

(C) 

(D) 

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Questions 4 to 5

The concentration of potassium ions inside a biological cell is at least twenty times higher than the outside. The resulting potential difference across the cell is important in several processes such as transmission of nerve impulses and maintaining the ion balance. A simple model for such a concentration cell involving a metal M is :

$M(s)\,|M^+(aq\,;\,0.05\text{ molar})\,||\,M^+(aq\,;\,1\text{ molar})\,|\,M(s)$

For the above electrolytic cell the magnitude of the cell potential $|E_{cell}| = 70$ mV.

Q4. For the above cell :-

(A) $E_{cell} < 0\,;\,\Delta G > 0$

(B) $E_{cell} > 0\,;\,\Delta G < 0$

(C) $E_{cell} < 0\,;\,\Delta G^0 > 0$

(D) $E_{cell} > 0\,;\,\Delta G^0 < 0$

Q5. If the 0.05 molar solution of M^+ is replaced by a 0.0025 molar M^+ solution, then the magnitude of the cell potential would be ?

(A) 35 mV (B) 70 mV

(C) 140 mV (D) 700 mV

Paragraph for Questions 6 to 8

Tollen's reagent is used for the detection of aldehyde when a solution of $AgNO_3$ is added to glucose with NH_4OH then gluconic acid is formed

$Ag^+ + e^- \longrightarrow Ag\,;\ E^0_{red} = 0.8$ V

$C_6H_{12}O_6 + H_2O \longrightarrow C_6H_{12}O_7$(Gluconic acid) $+ 2H^+ + 2e^-\ ;\ E^0_{red} = -0.05$ V

$Ag(NH_3)_2^+ + e^- \longrightarrow Ag(s) + 2NH_3\,;\ E^0 = -0.337$ V

[Use $2.303 \times \dfrac{RT}{F} = 0.0592$ and

$\dfrac{F}{RT} = 38.92$ at 298 K]

Q6. $2Ag^+ + C_6H_{12}O_6 + H_2O \longrightarrow 2Ag(s) + C_6H_{12}O_7 + 2H^+$

Find ln K of this reaction :

(A) 66.13 (B) 58.38

(C) 28.30 (D) 46.29

Q7. When ammonia is added to the solution, pH is raised to 11. Which half-cell reaction is affected by pH and by how much?

(A) E_{oxd} will increase by a factor of 0.65 from E^0_{oxd}

(B) E_{oxd} will decrease by a factor of 0.65 from E^0_{oxd}

(C) E_{red} will increase by a factor of 0.65 from E^0_{red}

(D) E_{red} will decrease by a factor of 0.65 from E^0_{red}

Q8. Ammonia is always is added in this reaction. Which of the following must be incorrect?

(A) NH_3 combines with Ag^+ to form a complex.

(B) $Ag(NH_3)_2^+$ is a weaker oxidising reagent than Ag^+.

(C) In absence of NH_3 silver salt of gluconic acid is formed.

(D) NH_3 has affected the standard reduction potential of glucose/gluconic acid electrode.

Paragraph for Question 9 to 11

Chemical reactions involve interaction of atoms and molecules. A large number of atoms/molecules (approximately 6.023×10^{23}) are present in a few grams of any chemical compound varying with their atomic/molecular masses. To handle such large numbers conveniently, the mole concept was introduced. This concept has implications in diverse areas such as analytical chemistry, biochemistry, electrochemistry and radiochemistry. The following example illustrates a typical case, involving chemical/ electrochemical reaction, which requires a clear understanding of the mole concept.

A 4.0 molar aqueous solution of NaCl is prepared and 500 mL of this solution is electrolysed. This leads to the evolution of chlorine gas at one of the electrodes (atomic mass : Na = 23, Hg = 200; 1 Faraday = 96500 coulombs)

Q9. The total number of moles of chlorine gas evolved is?

(A) 0.5 (B) 1.0 (C) 2.0 (D) 3.0

Q10. If the cathode is a Hg electrode, the maximum weight (g) of amalgam formed from this solution is ?

(A) 200 (B) 225

(C) 400 (D) 446

Q11. The total charge (coulombs) required for complete electrolysis is ?

(A) 24125 (B) 48250

(C) 96500 (D) 193000

Paragraph for Question 12 to 14

A sample of water from a large swimming pool has a resistance of 10000Ω at 25°C when placed in a certain conductance cell. When filled with 0.02 M KCl solution, the cell has a resistance of 100Ω at 25°C. 585 gm of NaCl were dissolved in the pool, which was throughly stirred. A sample of this solution gave a resistance of 8000 Ω.

Given : Molar conductance of NaCl at that concentration is $125\Omega^{-1}$ cm^2 mol^{-1} and molar conductivity of KCl at 0.02 M is 200 Ω^{-1} cm^2 mol^{-1}.

Q12. Cell constant (in cm^{-1}) of conductance cell is ?

(A) 4

(B) 0.4

(C) 4×10^{-2}

(D) 4×10^{-5}

Q13. Conductivity (Scm^{-1}) of H_2O is ?

(A) 4×10^{-2}

(B) 4×10^{-3}

(C) 4×10^{-5}

(D) None of these

Q14. Volume (in Litres) of water in the pool is ?

(A) 1.25×10^5

(B) 1250

(C) 12500

(D) 125

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q15.

Column-I	**Column-II**
(A) Conductance	(P) Cm^{-1}
(B) Specific conductance	(Q) Ohm^{-1} cm^2 mol^{-1}
(C) Cell constant	(R) Ohm^{-1}
(D) Equivalent conductance	(S) Ohm^{-1} cm^{-1}
(E) Molar conductance	(T) Ohm^{-1} cm^2 $equivalent^{-1}$

Q16.

Column-I	**Column-II**
(A) Cell constant	(P) $E^0_{cathode} + E^0_{anode}$
(B) Anode	(Q) ℓ/A
(C) Conductance	(R) Mass of product deposited by 1 coulomb of electricity.
(D) Electrochemical equivalent	(S) $(Resistance)^{-1}$
(E) E^0_{cell}	(U) Involve oxidation

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q17. Match the List :-

	Column I (Reduction process)		**Column II (Charge required)**
(a)	1 mol of MnO_4^- to Mn^{2+}	(p)	193000 C
(b)	1 mole of $Cr_2O_7^{2-}$ to Cr^{3+}	(q)	289500 C
(c)	1 mole of Sn^{4+} to Sn^{2+}	(r)	482500 C
(d)	1 mole of Al^{3+} to Al	(s)	579000 C

(A) a → p, b → r, c → q, d → s
(B) a → r, b → s, c → q, d → p
(C) a → s, b → q, c → p, d → r
(D) a → r, b → s, c → p, d → q

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

Q18. **Statement-I :** In electrolysis, the quantity of electricity needed for depositing 1 mole of silver is different from that required for 1 mole of copper.
Statement-II : The molecular weights of silver and copper are different.

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q19. For the redox reaction :
$Zn(s) + Cu^{2+}(0.1M) \rightarrow Zn^{2+}(1M) + Cu(s)$
taking place in a cell, E°_{Cell} is 1.10 volt. E_{Cell} for the cell will be $\left(2.303\frac{RT}{F} = 0.0591\right)$.

Q20. How much charge is present on 1 mole of Cu^{+2} ion in faraday.
(1 Faraday = 96500 coulomb)

Q21. Total charge in faraday present in 54 gm of Al^{3+} ion.

Q22. The value of μ^∞ for NH_4Cl, NaOH and NaCl are 129.8, 248.1 and 126.4 $ohm^{-1}\ cm^2\ mol^{-1}$ respectively. Calculate μ^∞ for NH_4OH solution.

Q23. Molar conductance of 1 M solution of weak acid HA is 20 $ohm^{-1}\ cm^2\ mol^{-1}$. Find % dissocaition of HA :

$$\left(\begin{matrix}\Lambda_m^\circ(H^+) = 350\ S\ cm^2\ mol^{-1} \\ \Lambda_m^\circ(A^-) = 50\ S\ cm^2\ mol^{-1}\end{matrix}\right)$$

SUBJECTIVE TYPE

Q24. Write cell representation for the following redox reaction,

$Sn^{2+}(aq) + 2Ag^{+}(aq) \rightarrow Sn^{4+}(aq) + 2Ag(s)$.

Q25. The 0.1M copper sulphate solution in which copper electrode is dipped at 25°C.

Calculate the reduction potential of copper electrode

[Given : $E^0\ Cu^{+2}\ /\ Cu = 0.34V$]

Q26. Calculate the equilibrium constant for the reaction $Fe^{2+} + Ce^{4+} \rightleftharpoons Fe^{3+} + Ce^{3+}$,

[Given: $E^0_{Ce^{4+}/Ce^{3+}} = 1.44V; E^0_{Fe^{3+}/Fe^{2+}} = 0.68V$]

Take $\frac{2.303\ RT}{F} = 0.06$ at 25°C, $\log 4.68 = 0.67$

Q27. How much electric charge is required to oxidise ?

(a) 1 mole of H_2O to O_2

(b) 1 mole of FeO to Fe_2O_3?

Q28. How long a current of 2A has to be passed through a solution of $AgNO_3$ to coat a metal surface of $80cm^2$ with 5μm thick layer? Density of silver = $10.8g/cm^3$.

Q29. How many litre of chlorine gas will be deposited by 100 amp. current flowing for 5 hours through melted NaCl.

Q30. The same current if passed through solution of silver nitrate and cupric salt connected in series. If the weight of silver diposited is 1.08 gm. Calculate the weight of copper deposited :-

Q31. The resistance of a 1N solution of salt is 50Ω. Calculate the equivalent conductance of the solution, if the two platinum electrodes in solution are 2.1cm apart and each having an area of $4.2cm^2$.

Q32. Calculate acid dissociation constant for 0.1 M HCOOH if its solution shows a resistance of 50 Ω when filled in a cell having separation between parallel electrodes 4 cm and cross section area of electrode 10 cm^2

Given : $\wedge^{\infty}_m [Ca(HCOO)_2] = 230\ Scm^2\ mol^{-1}$

$\wedge^{\infty}_m [CaCl_2] = 280\ Scm^2\ mol^{-1}$

$\wedge^{\infty}_m [HCl] = 425\ Scm^2\ mol^{-1}$

Q33. Calculate λ^{∞}_m of oxalic acid, given that :

$\lambda^{\infty}_{eq}\ Na_2C_2O_4 = 400\,\Omega^{-1}\ cm^2\ eq^{-1}$

$\lambda^{\infty}_m\ H_2SO_4 = 700\,\Omega^{-1}\ cm^2\ mole^{-1}$

$\lambda^{\infty}_{eq}\ Na_2SO_4 = 450\,\Omega^{-1}\ cm^2\ eq^{-1}$

Q34. The resistance of a 0.01 N solution of an electolyte was found to 210 ohm at 298 K using a conductivity cell with a cell constant of 0.88 cm^{-1}. Calculate specific conductance and equivalent conductance of solution.

Exercise - 2 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The electrode potential becomes equal to standard electrode potential when reactants and products concentration ratio is ?

(A) equal to 1

(B) greater than 1

(C) less than 1

(D) none of the above

Q2. The chemical reaction, $2AgCl(s) + H_2(g) \rightarrow 2HCl(aq) + 2Ag(s)$ taking place in a galvanic cell is represented by the notation

(A) $Pt|H_2(g)$, 1bar|1M KCl(aq)|AgCl(s)|Ag(s)

(B) $Pt(s)|H_2(g)$, 1bar|1M HCl(aq)||1M $Ag^+(aq)|Ag(s)$

(C) $Pt(s)|H_2(g)$, 1bar|1M HCl(aq)|AgCl(s)|Ag(s)

(D) $Pt(s)|H_2(g)$, 1bar|1M HCl(aq)|Ag(s)|AgCl(s)

Q3. The standard reduction potentials at 298 K for the following half reactions are given against each

$Zn^{2+}(aq) + 2e \square Zn(s)$; –0.762

$Cr^{3+}(aq) + 3e \square Cr(s)$; –0.740

$2H^+(aq) + 2e \square H_2(g)$; 0.00

$Fe^{3+}(aq) + e \square Fe^{2+}(aq)$; 0.770

Which is the strongest reducing agent

(A) Zn(s) (B) Cr(s)

(C) $H_2(g)$ (D) $Fe^{2+}(aq)$

Q4. The standard reduction potentials of Cu^{2+}, Zn^{2+}, Sn^{2+} and Ag^+ are 0.34, –0.76, – 0.14 and 0.80 V respectively. The storage that is possible without any reaction is for (under standard condition) :

(A) $CuSO_4$ solution in a zinc vessel

(B) $AgNO_3$ solution in a zinc vessel

(C) $AgNO_3$ solution in a tin vessel

(D) $CuSO_4$ solution in a silver vessel

Q5. The standard electrode potential for the following reaction is +1.33V.

What is the potential at pH = 2.0.

$Cr_2O_7^{2-}(aq\ 1M) + 14H^+(aq) + 6e^- \rightarrow 2Cr^{+3}(aq,1M) + 7H_2O\ (\ell)$

(A) 1.820 V (B) + 1.990 V

(C) + 1.608 V (D) + 1.0542 V

Q6. If $E^\circ_{Au^+/Au}$ is 1.69 volt and $E^\circ_{Au^{+3}/Au}$ is 1.40 volt then $E^\circ_{Au^+/Au^{+3}}$ will be ?

(A) 0.19

(B) 1.255 V

(C) –1.255 V

(D) 2.945 V

Q7. In the electrolytic cell, flow of electrons is from :

(A) Cathode to anode in solution

(B) Cathode to anode through external supply

(C) Cathode to anode through internal supply

(D) Anode to cathode through internal supply.

Q8. When an electric current is passed through a cell containing an electrolyte, positive ions move towards the cathode and negative ions towards the anode. What will happen if the cathode is pulled out of the solution?

(A) The positive ions will start moving towards the anode and negative ions will stop moving.

(B) The negative ions will continue to move towards the anode and the positive ions will stop moving.

(C) Both positive and negative ions will move towards the anode.

(D) None of these movements will take place.

Q9. On the basis of the electrochemical theory of aqueous corrosion, the reaction occurring at the cathode is ?

(A) $O_{2(g)} + 4H^+_{(aq)} + 4e^- \rightarrow 2H_2O_{(\ell)}$

(B) $Fe_{(s)} \rightarrow Fe^{2+}_{(aq)} + 2e^-$

(C) $Fe^{2+}_{(aq)} \rightarrow Fe^{3+}_{(aq)} + e^-$

(D) $H_{2(g)} + 2OH^-_{(aq)} \rightarrow 2H_2O_{(\ell)} + 2e^-$

Q10. Mark the false statement :

(A) A salt bridge is used to eliminate liquid junction potential.

(B) The Gibbs free energy change, ΔG is related with electromotive force (E), as $\Delta G = -nFE$

(C) Nernst equation for single electrode potential is $E = E^\circ - \frac{RT}{nF}\ell n a_{M^{n+}}$

(D) The efficiency of a hydrogen oxygen fuel cell is 23%

Q11. Deduce from the following E^o values of half cells, what combination of two half cells would result in a cell with the largest potential :-

(i) $A^{3-} \rightarrow A^{2-} + e$

(ii) $B^{2+} + e \rightarrow B^+$

$E^o = 1.5$ V

$E^o = -2.1$ V

(iii) $C^{2+} + e \rightarrow C^+$

(iv) $D \rightarrow D^{2+} + 2e$

$E^o = +0.5$ V

$E^o = -1.5$ V

(A) (i) and (iii) (B) (i) and (iv)

(C) (ii) and (iv) (D) (iii) and (iv)

Q12. If the solution of the $CuSO_4$ in which copper rod is immersed is diluted to 10 times, the electrode reduction potential. (T = 298 K) :-

(A) increases by 0.030 volt

(B) decreases by 0.030 volt

(C) increases by 0.059 volt

(D) decreases by 0.0059 volt

Q13. The cell Pt (H_2) (1 atm) $|H^+$ (pH = ?), I^- (a = 1) AgI(s), Ag has emf, $E_{298\,K} = 0$.

The electrode potential for the reaction

$AgI + e^- \rightarrow Ag + I^\ominus$ is –0.151 volt. Calculate the pH value ?

(A) 3.37 (B) 5.26

(C) 2.55 (D) 4.62

Q14. The standard e.m.f. of a cell, involving one electron change is found to be 0.591 V at 25°C. The equilibrium constant of the reaction is

(F = 96,500C mol^{-1} ; R = 8.314 JK^{-1} mol^{-1})

(A) 1.0×10^{10} (B) 1.0×10^5

(C) 1.0×10^1 (D) 1.0×10^{30}

Q15. The emf of the following galvanic cells :-

(a) $Zn \mid Zn^{+2}(1M) \parallel Cu^{+2}(1M) \mid Cu$

(b) $Zn \mid Zn^{+2}(0.1M) \parallel Cu^{+2}(1M) \mid Cu$

(c) $Zn \mid Zn^{+2}(1M) \parallel Cu^{+2}(0.1M) \mid Cu$

(d) $Zn \mid Zn^{+2}(0.1M) \parallel Cu^{+2}(0.1M) \mid Cu$

are represented by E_1, E_2, E_3 and E_4. Which of the following is true.

(A) $E_1 > E_2 > E_3 > E_4$

(B) $E_3 > E_2 > E_1 > E_4$

(C) $E_3 > E_1 = E_4 > E_2$

(D) $E_2 > E_1 = E_4 > E_3$

Q16. Consider the cell, $Cu|Cu^{+2}||Ag^+|Ag$. If the concentration of Cu^{+2} and Ag^+ ions becomes ten times, the emf of the cell :-

(A) Becomes 10 times

(B) Remains same

(C) Increases by 0.0295 V

(D) Decreases by 0.0295 V

Q17. Consider a voltaic cell based on these half-cells

$Ag^+(aq) + e^- \rightarrow Ag(s)$; $E^\circ = +0.80$ V

$Cd^{2+}(aq) + 2e^- \rightarrow Cd(s)$; $E^\circ = -0.40$ V

Identify the anode and give the voltage of the cell under standard conditions :-

(A) Ag ; $E^0_{cell} = 0.40$ V

(B) Ag ; $E^0_{cell} = 2.00$ V

(C) Cd ; $E^0_{cell} = 1.20$ V

(D) Cd ; $E^0_{cell} = 2.00$ V

Q18. The value of $E^{\circ}_{Zn/Zn^{2+}} = 0.76$V and that of $E^{\circ}_{Fe^{2+}/Fe} = -0.41$ V. The E°_{Cell} of the cell with net cell reaction, $Zn + Fe^{2+} \rightarrow Zn^{2+} + Fe$ is?

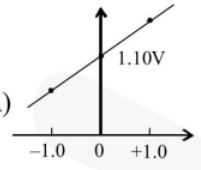
(A) –0.35 V (B) –1.17 V

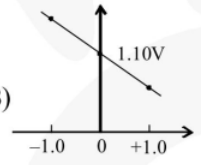
(C) +1.17 V (D) +0.35 V

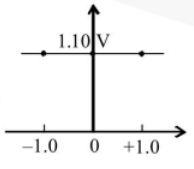
Q19. Which graph correct correlates E_{cell} as a function of concentrations for the cell (for different values of M and M') :

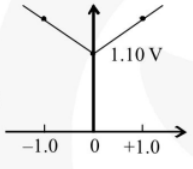
$Zn(s) + Cu^{2+}(M) \longrightarrow Zn^{2+}(M') + Cu(s)$; $E^{\circ}_{cell} = 1.10$ V

X-axis : $\log_{10}\frac{[Zn^{2+}]}{[Cu^{2+}]}$, Y-axis : E_{cell}

(A)


(B)


(C)


(D)


Q20. The standard reduction potentials of C^{2+}/C^{+} and C^{2+}/C are 0.339 V and 0.153 V respectively. The standard electrode potential of C^{+}/C half cell is?

(A) 0.525 V (B) 8.27 V

(C) 0.184 V (D) –0.033 V

Q21. Efficiency of the following cell is 84%

$A(s) + B^{+2}(aq.) \rightleftharpoons A^{+2}(aq.) + B(s)$

$\Delta H^{o} = -285$kJ

Then the standard electrode potential of the cell will be ?

(A) 1.10 V (B) 1.20 V

(C) 1.24 V (D) 2.40 V

Q22. Consider the cell $H_2(Pt)\underset{1atm}{\Big|}H_3O^{+}(aq)\underset{pH=5.5}{\Big|\Big|}Ag^{+}\underset{xM}{\Big|}$

Ag. The measured EMF of the cell is 1.023 V. What is the value of x ?

$E^0_{Ag^+, Ag} = +0.799$ V. [T = 25°C]

(A) 2×10^{-2} M

(B) 2×10^{-3} M

(C) 1.5×10^{-3} M

(D) 1.5×10^{-2} M

Q23. A standard hydrogen electrode has zero electrode potential because :

(A) Hydrogen is easiest to oxidize

(B) This electrode potential is assumed to be zero

(C) Hydrogen atom has only one electron

(D) Hydrogen is the lightest element

Q24. For the redox reaction :

$Zn(s) + Cu^{2+}(0.1M) \rightarrow Zn^{2+}(1M) + Cu(s)$ taking place in a cell,

E°_{Cell} is 1.10 volt. E_{Cell} for the cell will be

$\left(2.303\frac{RT}{F} = 0.0591\right)$

(A) 1.07 volt (B) 0.82 volt

(C) 2.14 volt (D) 1.80 volt

Q25. For the reaction ;

$2A_{(s)} + B^{2+}_{(aq)} \longrightarrow 2A^{+}_{(aq)} + B_{(s)}$,

Nernst equation for the EMF of the cell is ?

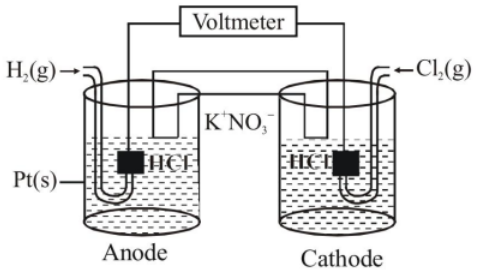
(A) $E_{cell} = E^{\circ}_{cell} - \frac{2.303\,RT}{F}\log\frac{[A^+]}{[B^{2+}]}$

(B) $E_{cell} = E^{\circ}_{cell} + \frac{2.303\,RT}{2F}\log\frac{[A^+]^2}{[B^{2+}]}$

(C) $E_{cell} = E^{\circ}_{cell} - \frac{2.303RT}{F}\log\frac{[A^+]}{\sqrt{[B^{2+}]}}$

(D) None

Q26. Consider the following Galvanic cell



By what value the cell potential change when concentration of ions in both anodic and cathodic compartments is increased upto 10 times at 298 K :-

(A) +0.0591 (B) –0.0591

(C) –0.1182 (D) 0

Q27. Starting with 250 ml of Au^{3+} solution 250 ml of Fe^{2+} solution, the following equilibrium is established.

$Au^{3+}(aq) + 3Fe^{2+}(aq) \rightleftharpoons 3Fe^{3+}(aq) + Au(s)$

At equilibrium, the equivalents of Au^{3+}, Fe^{2+}, Fe^{3+} and Au are x, y, z and w respectively. Then : -

(A) $K_C = \frac{Z^3}{6xy^3}$ (B) $K_C = \frac{3Z^3}{2xy^3}$

(C) $K_C = \frac{4Z^3}{9xy^3}$ (D) $K_C = \frac{8Z^3}{9xy^3}$

Q28. Given the data at 25°C,

$Ag_{(s)} + I^-_{(aq)} \rightarrow AgI_{(s)} + e^-$, $E° = 0.152V$

$Ag_{(s)} \rightarrow Ag^+_{(aq)} + e^-$, $E° = -0.800$ V

What is the value of log K_{sp} for AgI ?

(Where K_{sp} = solubility product)

$\left(2.303\frac{RT}{F} = 0.059V\right)$

(A) –8.12 (B) +8.612

(C) –37.83 (D) –16.13

Q29. $Pt\left|\frac{H_2}{(P_1)}\right|\frac{H^+}{(1M)}\Big\|\frac{H^+}{(1M)}\left|\frac{H_2}{(P_2)}\right|Pt$ (where P_1 and P_2 are pressures) cell reaction will be spontaneous if ?

(A) $P_1 = P_2$ (B) $P_1 > P_2$

(C) $P_2 > P_1$ (D) P_1 = 1 atm

Q30. Calculate cell potential at 298 K for following galvanic cell

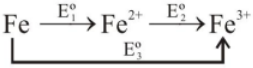
$Cd(s)|Cd^{+2}(aq.)(0.1M)\|H^+_{(aq)}(0.1M)|H_2(g, 0.5\,atm)|Pt$

$E°_{Cd^{+2}/Cd} = -0.40V$

(A) 0.38 (B) – 0.38

(C) 0.36 (D) – 0.36

Q31. Value of $E°_3$ may be calculated from :-



(A) $\frac{E°_1}{E°_2} + 1$ (B) $\frac{E°_2 + 2E°_1}{3}$

(C) $\frac{E°_1 + E°_2}{2}$ (D) $\frac{E°_1 + 2E°_2}{2}$

Q32. A cell is constituted as follows

$Pt\,|\,H_2\,(1\,atm)\,|\,HA_1\,|\,|\,HA_2\,|\,H_2\,(1\,atm)|\,Pt$

The pH of two acid solutions HA_1 and HA_2 are 5 and 3 respectively. The emf of the cell is :-

(A) 0.059 V (B) 0.0295 V

(C) 0.118 V (D) – 0.118 V

Q33. Calculate the quantity of electricity that would be required to reduce 12.3 gm of Nitrobenzene to aniline. If current efficiency is 50%.

(A) 57,900 coulomb (B) 115800 coulomb

(C) 17,3700 coulomb (D) 18950 coulomb

Q34. A solution of sodium sulphate in water is electrolysed using inert electrodes. The products at the cathode and anode are respectively :

(A) H_2, O_2 (B) O_2, H_2

(C) O_2, Na (D) O_2, SO_2

Q35. The Gibbs energy for the decomposition of Al_2O_3 at 500 °C is as follows :

$\frac{2}{3}Al_2O_3 \longrightarrow \frac{4}{3}Al + O_2$, $\Delta_rG = +966$ KJ mol^{-1}

The potential difference needed for electrolytic reduction of Al_2O_3 at 500 °C is at least :-

(A) 5.0 V (B) 4.5 V

(C) 3.0 V (D) 2.5 V

Q36. Which of the following statement(s) is(are) correct ?

(a) The electrolysis of molten NaOH liberates O_2 at the anode.

(b) The electrolysis of molten NaOH liberates O_2 at the cathode.

(c) Molten NaCl conducts electricity due to the presence of Na^+ and Cl^- ions.

(d) The electrolysis of molten KCl produces Cl_2 at the cathode.

(A) a, b (B) b, c

(C) a, c (D) c, d

Q37. A chemist wants to produce Cl_2 (g) from molten NaCl. How many grams could be produced if he uses a steady current of 2 ampere for 2.5 minutes ?

(A) 3.55 g (B) 1.775 g

(C) 0.110 g (D) 0.1775 g

Q38. The electric charge for electro deposition of 1 equivalent of a substance is?

(A) one ampere per second

(B) 96500 coulomb per second

(C) one ampere for one hour

(D) charge on one mole of electrons

Q39. Ethane gas is obtained in Kolbe's electrolysis of CH_3COONa according to the following reaction

$$2CH_3COO^- \longrightarrow \begin{matrix} CH_3 \\ | \\ CH_3 \end{matrix} + 2CO_2 + 2e^-$$

What volume of gas (ethane gas) at S.T.P. would be obtained by a current of 0.5 amp (80% efficient); if the current is passed for 965 min ?

(A) 1.8 L (B) 2.688 L

(C) 3.45 L (D) 11.2 L

Q40. During electrolysis of aq.NaCl, if 3mole of H_2O are electrolysed then how much charge is required if current efficiency is 75% -

(A) 1 F (B) 2 F

(C) 4 F (D) 8 F

Q41. The same amount of current was passed for same time through molten Al_2O_3 and molten NaCl. If 1.8 gm of aluminium is liberated in one cell the amount of sodium liberated in the other is? (Al → 27, Na → 23)

(A) 4.6 gm (B) 2.3 gm

(C) 8.0 gm (D) 3.6 gm

Q42. The number of Faraday required to generate 1 g atom of Mg from $MgCl_2$ is ?

(A) 1 (B) 2 (C) 3 (D) 4

Q43. Electrolytic conduction differs from metallic conduction in that in the case of electrolytic conduction

(A) The resistance increases with increasing temperature.

(B) The resistance decreases with increasing temperature.

(C) The flow of current does not generate heat.

(D) The resistance is independent of the length of the conductor.

Q44. In an electroplating experiment m gm of silver is deposited, when 4 amperes of current flows for 2 minutes. The amount (in gms) of silver deposited by 6 amperes of current flowing for 40 sec. will be?

(A) 4m (B) m/2

(C) m/4 (D) 2m

Q45. The resistance of 0.0025 M solution of K_2SO_4 is 326 ohm. The specific conductance of the solution is ?

(A) 4.997×10^{-4} (B) 5.997×10^{-7}

(C) 6.997×10^{-4} (D) Unpredictable

Q46. Resistance of a conductivity cell filled with a solution of an electrolyte of concentration 0.1M is 100Ω. The conductivity of this solution is 1.29 Sm^{-1}. Resistance of the same cell when filled with 0.02M of the same solution is 520Ω. The molar conductivity of 0.02M solution of the electrolyte will be?

(A) 124×10^{-4} Sm^2 mol^{-1}

(B) 1240×10^{-4} Sm^2 mol^{-1}

(C) 1.24×10^{4} Sm^2 mol^{-1}

(D) 12.4×10^{-4} Sm^2 mol^{-1}

Q47. At 291 K, the molar conductivity at infinite dilution of NH_4Cl, NaOH, NaCl are 129.8, 217.4, 108.9 ohm^{-1} cm^2 mol^{-1} respectively. If the molar conductivity of centinormal solution of NH_4OH is 9.33 ohm^{-1} cm^2 mol^{-1}. What is the percentage dissociation of NH_4OH at this dilution ?

(A) 0.392 (B) 39.2

(C) 3.91 (D) 0.039

Q48. $\lambda_{CH_3COOH} = 20$ ohm^{-1} cm^2 eq^{-1} and $\lambda^{\infty}_{CH_3COOH} =$ 400 ohm^{-1} cm^2 eq^{-1} calculate the pH of 1M solution of this acid :-

(A) 1.3 (B) 0

(C) 1.7 (D) 4

Q49. The conductivity of a saturated solution of $BaSO_4$ is 3.06×10^{-6} ohm^{-1} cm^{-1} and its molar conductance is 1.53 ohm^{-1} cm^2 mol^{-1}. The K_{sp} of $BaSO_4$ will be ?

(A) 4×10^{-12} (B) 2.5×10^{-9}

(C) 2.5×10^{-13} (D) 4×10^{-6}

Q50. The conductivity of a 0.01N solution is found to be 0.005 $ohm^{-1}cm^{-1}$. The equivalent conductivity of the solution will be ?

(A) 500 $ohm^{-1}cm^2eq^{-1}$

(B) 5×10^{-2} $ohm^{-1}cm^2eq^{-1}$

(C) 0.5 $ohm^{-1}cm^2eq^{-1}$

(D) 2 $ohm^{-1}cm^2eq^{-1}$

Q51. The correct order of equivalent conductance at infinite dilution of LiCl, NaCl and KCl is?

(A) LiCl > NaCl > KCl

(B) KCl > NaCl > LiCl

(C) NaCl > KCl > LiCl

(D) LiCl > KCl > NaCl

Q52. The potential of the Daniel cell, $Zn\left|\begin{matrix}ZnSO_4\\(1M)\end{matrix}\right|\left|\begin{matrix}CuSO_4\\(1M)\end{matrix}\right|Cu$ was reported by Buckbee, Surdzial and Metz as $E^\circ = 1.1028 - 0.641 \times 10^{-3} T + 0.72 \times 10^{-5} T^2$, where T is the celsius temperature. Calculate ΔS° for the cell reaction at 25°C :-

(A) –45.32 EU (B) –34.52 EU

(C) –25.43 EU (D) –54.23 EU

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The EMF of the following cell is 0.22 volt.

$Ag(s) \mid AgCl(s) \mid KCl\ (1M) \mid H^+(1M) \mid H_2(g)\ (1atm)\ ;\ Pt(s).$

Which of the following will decrease the EMF of cell.

(A) increasing pressure of $H_2(g)$ from 1 atm to 2 atm

(B) increasing Cl^- concentration in Anodic compartment

(C) increasing H^+ concentration in cathodic compartment

(D) decreasing KCl concentration in Anodic compartment.

Q2. Which of the following arrangement will produce oxygen at anode during electrolysis ?

(A) Dilute H_2SO_4 solution with Cu electrodes.

(B) Dilute H_2SO_4 solution with inert electrodes.

(C) Fused NaOH with inert electrodes.

(D) Dilute NaCl solution with inert electrodes.

Q3. The following data are obtained in an experiment

Solution	**Conductivity**
0.01 M HA	$3.8 \times 10^{-5}\ \Omega^{-1}\ cm^{-1}$
100 ml of 0.01 M HA + 1 mmol NaOH	$80.0 \times 10^{-5}\ \Omega^{-1}\ cm^{-1}$

Calculate

Given : $\lambda_m^{\infty}\ (Na^+) = 50\Omega^{-1}\ cm^2\ mol^{-1}$, $\lambda_m^{\infty}\ (HA) = 350\Omega^{-1}\ cm^2\ mol^{-1}$

(A) Λ_m (NaA) is $80\Omega^{-1}\ cm^2\ mol^{-1}$

(B) $\Lambda_m\ (A^-)$ is $30\Omega^{-1}\ cm^2\ mol^{-1}$

(C) Λ_m (HA) is $3.8\ \Omega^{-1}\ cm^2\ mol^{-1}$

(D) degree of dissociation of 0.01 M HA is 0.01

Q4. Equivalent conductance of $BaCl_2$, H_2SO_4 & HCl at infinite dilution are $A_{\infty}^{\ 1}$, $A_{\infty}^{\ 2}$ & $A_{\infty}^{\ 3}$ respectively. Equivalent conductance of $BaSO_4$ solution is :

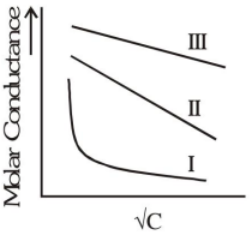
(A) $A_{\infty}^{\ 1} + A_{\infty}^{\ 2} - 2\ A_{\infty}^{\ 3}$

(B) $A_{\infty}^{\ 1} + A_{\infty}^{\ 2} + A_{\infty}^{\ 3}$

(C) $A_{\infty}^{\ 1} + A_{\infty}^{\ 2} - A_{\infty}^{\ 3}$

(D) $A_{\infty}^{\ 1} - A_{\infty}^{\ 2} + A_{\infty}^{\ 3}$

Q5. A graph was poltted between the molar conductance of various electrolytes (HCl, KCl and CH_3COOH) and root of their concentrations in mole per litre.



Which of the following is correct match?

(A) I (CH_3COOH) ; II (KCl) ; III (HCl)
(B) I (HCl) ; II (KCl) ; III (CH_3COOH)
(C) I (CH_3COOH) ; II (HCl) ; III (KCl)
(D) I (KCl) ; II (CH_3COOH) ; III (HCl)

Q6. During discharging of lead storage battery, which of the following is/are true ?

(A) H_2SO_4 is produced
(B) H_2O is consumed
(C) $PbSO_4$ is formed at both electrodes
(D) Density of electrolytic solution decreases

Q7. What a lead storage is discharged :-

(A) $PbSO_4$ is formed
(B) Pb is formed
(C) SO_2 is consumed
(D) H_2SO_4 is consumed

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question 8 & 9

Redox reactions play a pivoted role in chemistry and biology. The values of standard redox potential (E°) of two half-cell reactions decide which way the reaction is expected to proceed. A simple example is a Daniel cell in which zinc goes into solution and copper gets deposited. Given below are a set of half-cell reactions (acidic medium) along with their E° (V with respect to normal hydrogen electrode) values. Using this data obtain the correct explanations to Questions 16-17.

$I_2 + 2e^- \rightarrow 2I^-$	E° = 0.54
$Cl_2 + 2e^- \rightarrow 2Cl^-$	E° = 1.36
$Mn^{3+} + e^- \rightarrow Mn^{2+}$	E° = 1.50
$Fe^{3+} + e^- \rightarrow Fe^{2+}$	E° = 0.77
$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$	E° = 1.23

Q8. Among the following, identify the correct statement.

(A) Chloride ion is oxidised by O_2
(B) Fe^{2+} is oxidised by iodine
(C) Iodine ion is oxidised by chlorine
(D) Mn^{2+} is oxidised by chlorine

Q9. While Fe^{3+} is stable, Mn^{3+} is not stable in acid solution because

(A) O_2 oxidises Mn^{2+} to Mn^{3+}

(B) O_2 oxidises both Mn^{2+} to Mn^{3+} and Fe^{2+} to Fe^{3+}

(C) Fe^{3+} oxidises H_2O to O_2

(D) Mn^{3+} oxidises H_2O to O_2

Paragraph for Question 10 to 12

A half cell is prepared by $K_2Cr_2O_7$ in a buffer solution of pH = 1. Concentration of $K_2Cr_2O_7$ is 1 M. To 3 litre of this solution 570 gm of $SnCl_2$ is added which is oxidised completely to $SnCl_4$.

Given : $E^{\circ}_{Cr_2O_7^{-2}/Cr^{+3},H^+} = 1.33$ V ; $\frac{2.303}{F}RT = 0.06$;

Atomic of mass Sn = 119 ; $E^{\circ}_{Sn^{+4}/Sn^{+2}} = 0.15$

Q10. Number of moles of Cr^{+3} formed are?

(A) 2 (B) 6 (C) 4 (D) 3

Q11. Half cell potential $E_{Cr_2O_7^{-2}/Cr^{+3}}$ after the reaction of $SnCl_2$ is ?

(A) 1.187 V (B) 1.191 V (C) 1.285 V (D) 1.537 V

Q12. Emf of the cell $Pt \mid \underset{(0.1\ M)}{Sn^{+2}}, \underset{(0.2\ M)}{Sn^{+4}}, \underset{(1M)}{H^+} \parallel \underset{(0.2\ M)}{Cr_2O_7^{2-}}, \underset{(1\ M)}{Cr^{+3}}, \underset{(1\ M)}{H^+} \mid Pt$

(A) – 1.18 V (B) 1.164 V (C) 1.18 V (D) None of these

Paragraph for Question 13 & 14

Copper reduces NO_3^- into NO and NO_2 depending upon conc. of HNO_3 in solution.

Assuming $[Cu^{2+}] = 0.1$ M, and $P_{NO} = P_{NO_2} = 10^{-3}$ atm and using given data answer the following questions:

$E^{o}_{Cu^{2+}/Cu} = +0.34$ volt

$E^{o}_{NO_3^-/NO} = +0.96$ volt

$E^{o}_{NO_3^-/NO_2} = +0.79$ volt

at 298 K $\frac{RT}{F}(2.303) = 0.06$ volt

Q13. E_{Cell} for reduction of $NO_3^- \longrightarrow NO$ by Cu(s), when $[HNO_3] = 1M$ is [At T = 298]

(A) ~ 0.61 (B) ~ 0.71 (C) ~ 0.51 (D) ~ 0.81

Q14. At what HNO_3 concentration thermodynamic tendency for reduction of NO_3^- into NO and NO_2 by copper is same?

(A) $10^{1.23}$ M (B) $10^{0.56}$ M (C) $10^{0.66}$ M (D) $10^{0.12}$ M

Paragraph for Question 15 to 17

Accidentally chewing on a stray fragment of aluminium foil can causes a sharp tooth pain if the aluminium comes in contact with an amalgam filling. The filling, an alloy of silver, tin and mercury, acts as the cathode of a tiny galvanic cell, the aluminium behaves as the anode, and saliva serves as the electrolyte. When the aluminium and the filling come in contact, an electric current passage from the aluminium to the filling which is sensed by a nerve in the tooth. Aluminium is oxidized at the anode, and O_2 gas is reduced to water at the cathode.

$E^\circ_{Al^{3+}/Al} = -1.66$ V, $E^\circ_{O_2,H^+/H_2O} = 1.23$ V

Q15. Net reaction taking place when amalgam is in contact with aluminium foil :

(A) $Al + O_2 + OH^- \rightarrow Al(OH^-) + H_2O$ (B) $4Al + 3O_2 + 12H^+ \rightarrow 4Al^{3+} + 6H_2O$

(C) $4Al + 3O_2 \rightarrow 4Al_2O_3$ (D) $2H_2 + O_2 \rightarrow 2H_2O$

Q16. Standard E.M.F. experienced by the person with dental filling is :

(A) +2.89 V (B) –2.89 V (C) –0.93 V (D) +0.43 V

Q17. The standard reduction potential of the reaction, $H_2O + e^- \rightarrow \frac{1}{2}H_2 + OH^-$ at 298 K is ?

(A) $E^\circ = \frac{RT}{F}\ln K_w$ (B) $E^\circ = \frac{RT}{F}\ln\left[P_{H_2}\right]^{1/2}[OH^-]$

(C) $E^\circ = \frac{RT}{F}\ln\frac{\left[P_{H_2}\right]^{1/2}}{[H^+]}$ (D) $E^\circ = -\frac{RT}{F}\ln K_w$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q18.

Column-I	**Column-II**
(A) Electrolytic cell	(P) $-\Delta G^\circ$
(B) nFE^0_{Cell}	(Q) Concentration cell
(C) $E_{cell} = \frac{0.059}{n}\log\frac{C_{cathode}}{C_{anode}}$	(R) 96500 Coulombs
(D) Diffusion of ions	(S) Device converting electrical energy into chemical energy
(E) 1 Faraday	(U) Salt bridge

Q19.

Column I	Column II (Electrolysis product using inert electrode)
(A) Dilute solution of HCl	(P) O_2 evolved at anode
(B) Dilute solution of NaCl	(Q) H_2 evolved at cathode
(C) Concentrated solution of NaCl	(R) Cl_2 evolved at anode
(D) $AgNO_3$ solution	(S) Ag deposition at cathode

MATCHING LIST TYPE

Each question has TWO (02) ***matching lists: LIST I and LIST II****. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q20 .

	Column-I		Column-II
a	Kohlrausch law	p	$\frac{\wedge_m^c}{\wedge_m^o}$
b	Molar conductance, $\wedge_m$	q	$\frac{1}{R}\times\frac{\ell}{A}$
c	Specific conductance (k)	r	$\wedge_m^o$ of $Ca_3(PO_4)_2$ $= 3(\wedge_m^o)_{Ca^{+2}} + 2(\wedge_m^o)_{PO_4^{-3}}$
d	Degree of ionization of weak electrolyte	s	$\frac{K\times 1000}{M}$

Which of following option show correct matches ?

(A) (a–r), (b–p), (c–q), (d–s)
(B) (a–s), (b–p), (c–q), (d–r)
(C) (a–r), (b–s), (c–q), (d–p)
(D) (a–p), (b–s); (c–q); (d–r)

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

Q21. **Statement-I :** Absolute value of E^0_{red} of an electrode cannot be determined.
Statement-II : Neither oxidation nor reduction can take place alone.

Q22. **Statement-I :** Molar conductivity of a weak electrolyte at infinite dilution cannot be determined experimentally.
Statement-II : Kohlrausch law help to find the molar conductivity of a weak electrolyte at infinite dilution.

SUBJECTIVE TYPE

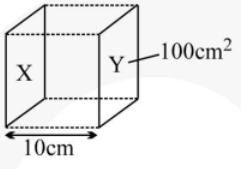
Q23. The measured e.m.f. at 25°C for the cell reaction,
$Zn\ (s) + Cu^{2+}_{(aq)}\ (1.0M) \longrightarrow Cu\ (s) + Zn^{2+}_{(aq)}\ (0.1\ M)$ is 1.3 volt, Calculate E° for the cell reaction.

Q24. Calculate E^0 and E for the cell $Sn|Sn^{2+}\ (1M) \parallel Pb^{2+}(10^{-3}M) \mid Pb$, $E^0\ (Sn^{2+}|\ Sn) = -0.14V$, $E^0\ (Pb^{2+}|\ Pb) = -0.13V$. Is cell representation correct?

Q25. The EMF of the cell
$Cr \mid Cr^{+3}\ (0.1M) \parallel Fe^{+2}\ (0.01M) \mid Fe$
(Given : $E°\ Cr^{+3} \mid Cr = -0.75V$
$E°\ Fe^{+2} \mid Fe = -0.45\ V$)

Q26. Exactly 0.4 faraday electric charge is passed through three electrolytic cells in series, first containing $AgNO_3$, second $CuSO_4$ and third $FeCl_3$ solution. How many gram of each metal will be deposited assuming only cathodic reaction in each cell.

Q27. The conductance of a salt solution (AB) measured by two parallel electrodes of area 100 cm^2 separated by 10 cm was found to be 0.0001 Ω^{-1}. If volume enclosed between two electrode contain 0.5 mole of salt, what is the molar conductivity(Scm^2mol^{-1}) of salt at same conc.



Q28. The equivalent conductivities of acetic acid at 298 K at the concentrations of 0.1 M and 0.001M are 5.20 and 49.2 S cm^2 $eq.^{-1}$ respectively. Calculate the degree of dissociation of acetic acid at these concentrations. Given that, $\lambda^\infty(H^+)$ and $\lambda^\infty(CH_3COO^-)$ are 349.8 and 40.9 $ohm^{-1}\ cm^2\ mol^{-1}$ respectively.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
C	A	D	C	C	C	B	C	D	C
11	12	13	14	15	16	17	18	19	20
B	A	B	B	C	C	B	B	C	A
21	22	23	24	25	26	27	28	29	30
B	A	B	D	C	C	D	C	C	A
31	32	33	34	35	36	37	38	39	40
A	D	C	B	C	D	C	B	D	B
41	42	43	44	45	46				
B	A	D	D	C	D				

Ex-1A

1. A,D **2.** A, B **3.** A **4.** B **5.** C

6. B **7.** A **8.** D **9.** B **10.** D

11. D **12.** B **13.** C **14.** A

15. A - (R), B - (S), C - (P), D - (T), E - (Q)

16. A - (Q), B - (U), C - (S), D - (R) , E - (P)

17. D **18.** B **19.** 1.07 **20.** 2 **21.** 6

22. 251.5 **23.** 5 **25.** 0.31 Volts **26.** 4.64×10^{12}

27. (a) 1.93×10^5 C ,(b) 96500C **28.** 193 sec **29.** 208.9 lit. **30.** 0.3175 gm

31. 10 S cm^2/ equivalent **32.** 5×10^{-3} M **33.** 600 S cm^2 mol^{-1}

34. Λ_{eq} = 419 mho cm^2 eq^{-1} ; k = 4.19×10^{-3} mho cm^{-1}

Ex-2

1	2	3	4	5	6	7	8	9	10
A	C	A	D	D	C	C	D	A	D
11	12	13	14	15	16	17	18	19	20
C	B	C	A	D	C	C	D	B	D
21	22	23	24	25	26	27	28	29	30
C	A	B	A	C	C	B	D	B	A
31	32	33	34	35	36	37	38	39	40
B	C	B	A	D	C	C	D	B	C
41	42	43	44	45	46	47	48	49	50
A	B	B	B	D	A	C	A	D	A
51	52								
B	D								

Ex-2A

1. A,D **2.** B,C,D **3.** A,B,C,D **4.** C **5.** A

6. C,D **7.** A,D **8.** C **9.** D **10.** A

11. B **12.** B **13.** B **14.** C **15.** A

16. A **17.** A

18. A - (S), B - (P), C - (Q), D - (U), E - (R)

19. (A) → P, Q ; (B) → P, Q ; (C) → Q, R, ; (D) → P,S

20. C **21.** A **22.** B **23.** 1.27 V **24.** – 0.07 V

25. 0.26 Volt **26.** 43.2 g, 12.7 g, 7.47 g **27.** 0.02 S cm^2 mol^{-1}

28. (i) 390.7 Ohm^{-1} cm^2 $eq.^{-1}$, (ii) 1.3%, 12.5%